



Mutual Trim+

Onshape's mutual trim, with the side to keep on each surface named by a reference instead of by a flip -- so the trim keeps the right corner whatever the surfaces' orientations.

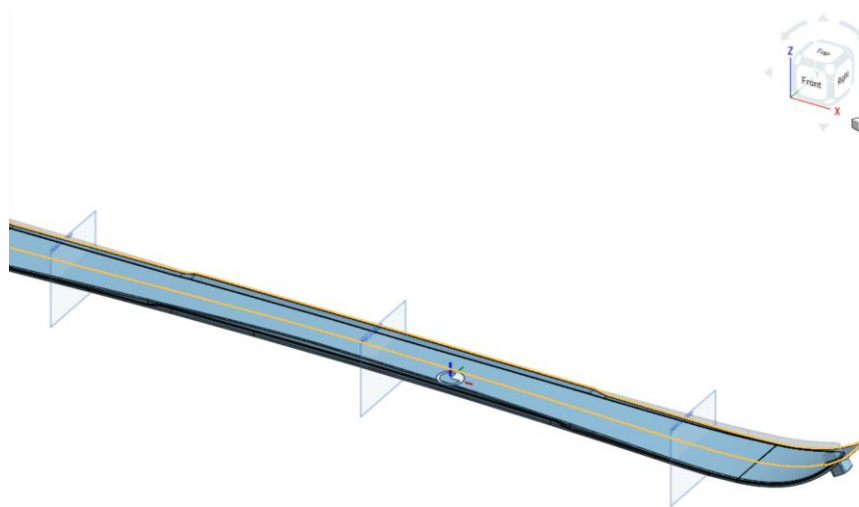


What it does

Two surfaces trim each other along their intersection and each keeps one side. The built-in picks that side with a 'keep opposite side' box per surface, tied to each surface's normal -- which flips when a loft, offset or mirror upstream changes. Mutual Trim+ keeps, on each surface, the side a reference is on.

Use it when

- Two surfaces meet at a corner you want to keep (a top and a wall)
- Either surface's orientation may change upstream
- A fillet or thicken will follow the trim edge



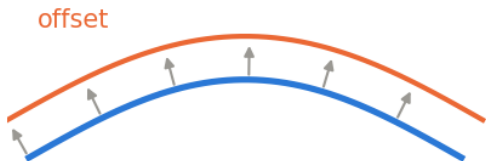
Mutual Trim+ open on the ski: unfilleted top x SW rout wall.



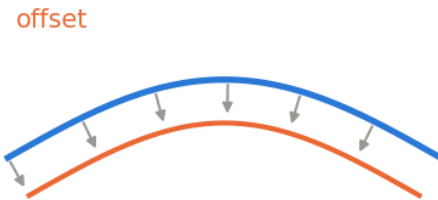
Why a reference instead of a flip

A flip names a side relative to a normal; a reference names it in the model

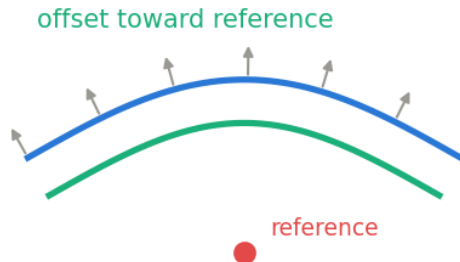
Built-in, normal up:
'offset along normal' goes up



Same model after an upstream change
flips the normal: now it goes DOWN



Offset+ 'toward reference':
goes toward the point, either way



A flip is about a normal

'Front', 'opposite direction', 'Thickness 1'
all mean: relative to the normal -- and
normals change freely upstream.

A reference is about the model

Pick geometry on the side you mean. The
choice survives any change of
orientation.

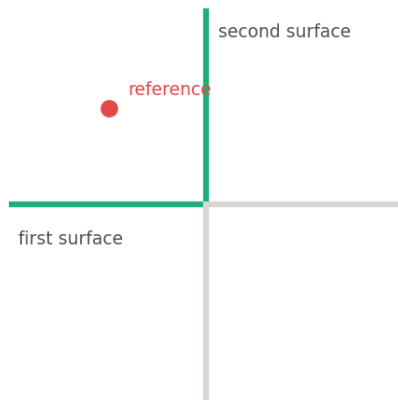
Drop-in

With no reference, the feature behaves
exactly like the built-in.

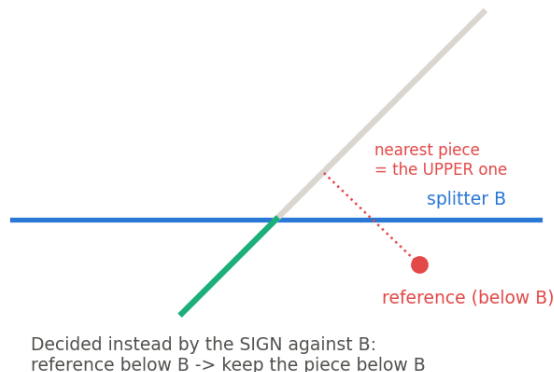


How each side is decided

Each surface keeps the side the reference is on
(green kept, grey removed)



Why 'nearest piece' fails on a leaning wall (correction 28)



Against the OTHER surface

Each surface is cut in two by the other one. The side kept is the one on the reference's side of that other surface (signed distance).

Not by nearness

On a leaning wall the nearest piece can be the wrong one (right). Distance is only a fallback when the pieces do not straddle the other surface.

Per surface

Each surface has its own 'keep side nearest reference' arrow: on = the reference's side, off = the other side.



The dialog

Mutual Trim+ unfilleted t... ☒ ☐

1 First surface
Unfilleted_Top_Surf x ↗

3 Second surface
SW_Rout x ↗

5 Keep side reference
Better_Ref x ⌚

6 ☒ Merge

7 > Debug

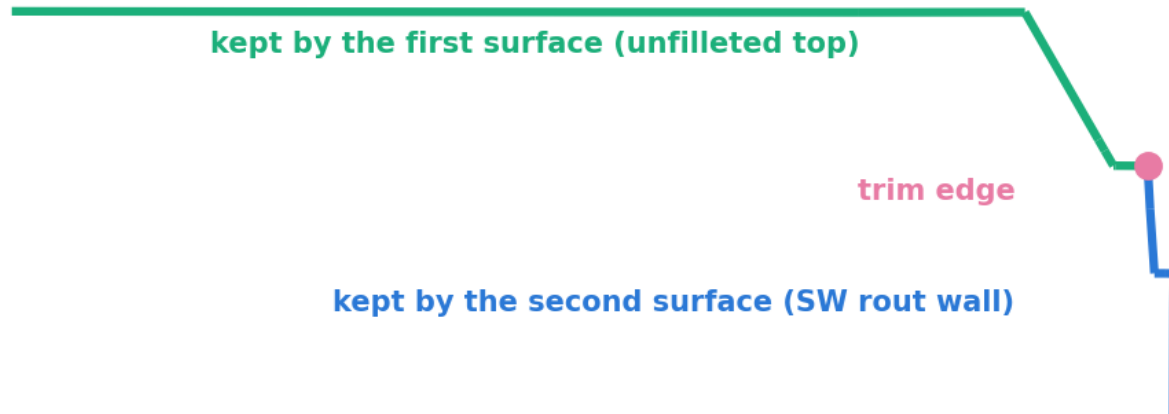
Final

#	Parameter	What it means
1	First surface	One surface body.
2	Keep side nearest reference (first)	The arrow by the first surface. On: keep its side nearer the reference (the inside). Off: the farther side.
3	Second surface	The other surface body.
4	Keep side nearest reference (second)	The same for the second surface.
5	Keep side reference	Geometry on the keep side of BOTH surfaces -- best a mate connector or vertex in the corner you keep. Empty = the built-in.
6	Merge	Union the two trimmed surfaces into one body (default on).
7	Debug	Print side distances: which side each surface kept, and whether by side (normal case) or by distance (fallback).



Example: top x sidewall rout

Unfilleted top x SW rout, trimmed: section at x 885 (looking along the ski)



Unfilleted_Top_Surf trimmed with SW_Rout, reference Better_Ref, merged (topsheet_surf studio)

Each surface keeps the part on the reference's side of the other: the top keeps its inside (green), the rout wall keeps the part below the top (blue). The published trimEdges (magenta) is the intersection along which the two were trimmed -- the edge a fillet wants.



Example: the whole trim edge

The trimmed result (plan view): top kept by the first surface, rout wall kept by the second, trim edge in magenta



width exaggerated x3

The same result from above: keptFaces1 (green) and the trim edge all round the ski (magenta)

keptFaces1 / keptFaces2 tell the two surfaces apart even after Merge makes them one body, so a later feature can offset or thicken just the top, or just the wall.



Outputs and messages

Published for Extract variables

Key	What it holds
output	The trimmed surfaces (one body when merged).
trimEdges	The intersection edges the trim left -- where to fillet. Tracked through the merge.
keptFaces1	The faces the first surface kept (still told apart after the merge).
keptFaces2	The faces the second surface kept.
trimEdgeCount	How many trim edges.

Messages

	Message / what to do
Info	boolean union no-op The surfaces do not intersect; nothing was trimmed.
Error	The keep side reference cannot be placed on either side of the first/second surface The pieces do not straddle the other surface and the reference is equally far from both. Pick a closer reference inside the corner to keep.



Tips and limits



Put the reference in the corner you keep

A mate connector just inside the kept corner (e.g. at MRS) -- not a far plane: a far plane's closest point can land at the tip, where the surfaces have turned.



Fillet along trimEdges

Publish trimEdges with Extract variables; it survives upstream edits that renumber edges.



Outputs keep their input's orientation

The reference picks the side, not the normal. Built-ins downstream that follow the normal (Thicken, Offset surface, Move face) need Orient to reference first -- or use the reference-side feature instead.



One orientation after Merge

Merged, the two halves are one body with one orientation.



Where it lives

Document	Reference_Side_Features (22764764a00a7f607dbc1c4d)
Tab	mutual_trim_plus (uses reference_side_utils)
Tests	Reference side tests studio: M1-M2 - devtools/onshape/check_reference_side_tests.py (34/34)
Examples	topsheet_surf studio (the user's derive of Design_Master topsheet and sidewall bodies)
Explainer	docs/explainers/reference_side/reference_side_explained.md, section 2.1
Related	Split+, Mutual Trim+, Offset+, Thicken+, Orient to reference (same reference rule); Extract variables